

JEE MAIN REVISION MASTER: PRACTICAL ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 111 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Practical Organic Chemistry — Applied Cases, Reactions & Traps	78 PYQs	Applied Topics
Very High	Practical Organic Chemistry — Standard Calculations & Problem Archetypes	37 PYQs	Quantitative Analysis
Very High	Practical Organic Chemistry — Core Principles & Governing Laws	19 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Practical Organic Chemistry — Applied Cases, Reactions & Traps	$\text{CuSO}_4 \cdot x\text{H}_2\text{O} + \text{NaSCN} + 2\text{Na} \rightarrow \text{NaCN} + \text{Na}_2\text{S} + \text{KMnO}_4$
Practical Organic Chemistry — Standard Calculations & Problem Archetypes	Moles of $\text{N}_2 = \frac{\text{Mass of N}_2}{\text{Molar Mass of N}_2}$ $\text{Moles of N}_2 = \frac{0.4\text{g AgBr}}{188} \times 0.4$
Practical Organic Chemistry — Core Principles & Governing Laws	$\text{C}_x\text{H}_y\text{O}_z + \left(x + \frac{y}{4} - \frac{z}{2}\right)\text{O}_2 \rightarrow x\text{CO}_2 + \frac{y}{2}\text{H}_2\text{O}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Practical Organic Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 78 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.481..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Identify the correct statements : A. Hydrated salts can be used as primary standard. B. Primary standard should not undergo any reaction with..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "Consider the following reactions sequence When the product (P) is subjected to Carius analysis using AgNO_3 , 1.0 g of the product (P) will ..." [Ans: 1]

3.2 Practical Organic Chemistry — Standard Calculations & Problem Archetypes (Very High Yield — 37 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.481..." [Ans: B]
- [2026 Evening Shift] Q#2 (MCQ): "Complete combustion of X g of an organic compound gave 0.25 g of CO_2 and 0.12 g of H_2O . If the % of carbon is 25% and of ..." [Ans: A]
- [2026 Evening Shift] Q#2 (MCQ): "When 1 g of compound (X) is subjected to Kjeldahl's method for estimation of nitrogen, 15 mL 1 M H_2SO_4 was neutralized by ammonia ..." [Ans: N/A]

3.3 Practical Organic Chemistry — Core Principles & Governing Laws (Very High Yield — 19 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Practical Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Practical Organic Chemistry PYQs.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in al..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Amongst the following, the total number of compounds soluble in aqueous NaOH at room temperature is" **[Ans: N/A]**
- **[2026 Morning Shift] Q#3 (MCQ):** "Given below are two statements : Statement (I) : 1,2,3-Trihydroxypropane can be separated from water by simple distillation. Statement (II) ..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 28th January Evening Shift (Q#1)

A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is _____. (Given Molar mass in g mol⁻¹ S: 32, BaSO₄ 4 : 233)

- (A) 63.15%
- (B) 21.05% **[Correct]**
- (C) 48.24%
- (D) 42.10%

Correct Answer: B

Solution Analysis: Mass of compound taken = 0.314g Mass of BaSO₄ obtained = 0.4813g In BaSO₄, sulphur present per mole = 32g Molar mass of BaSO₄ = 233g mol⁻¹ So, mass of sulphur in 0.4813g of BaSO₄ is:
 $m(S) = 0.4813 \times (32 / 233)$
 $m(S) = 0.4813 \times (32 / 233) = 0.06608$ g (approx.) Percentage of sulphur in the compound:
 $\%S = (0.06608 / 0.314) \times 100 \approx 21.05\%$ Correct option: B) 21.05%

Benchmark Problem 2: JEE Main 2024 (Online) 30th January Morning Shift (Q#3)

On a thin layer chromatographic plate, an organic compound moved by 3.5cm, while the solvent moved by 5cm. The retardation factor of the organic compound is _____ $\times 10^{-1}$.

Correct Answer: 7

Solution Analysis: Retardation factor = (Distance travelled by sample/organic compound / Distance travelled by solvent)
 $= (3.5 / 5) = 7 \times 10^{-1}$